

Open Source as Institutional Infrastructure

Lessons from the UC OSPO Network

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The invisible layer



Research data workflows run on open source:

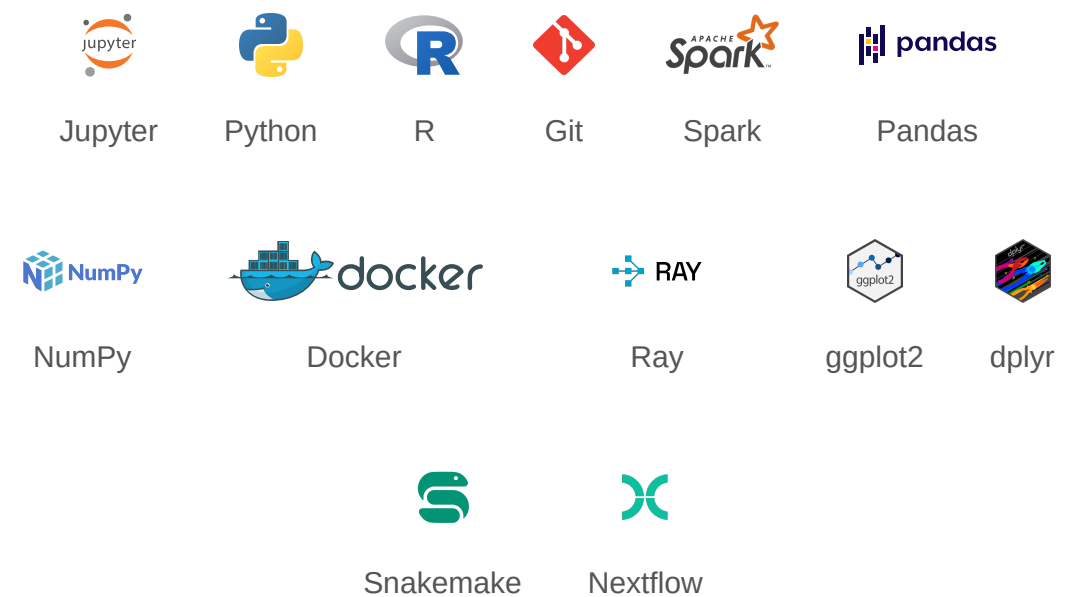
- Data collection, cleaning, analysis, visualization
- Reproducibility tooling: containers, workflows, version control
- Repositories and scholarly infrastructure

92% of researchers use software; **69%** say their research would not be practical without it.

The people keeping it running are increasingly called **Research Software Engineers (RSEs)**, a growing professional community with no standard institutional home.

If every organization had to replace open source with proprietary equivalents, the cost would exceed **\$8.8 trillion**.

Hettrick et al. (SSIJ, 2014/2022) · Hoffmann, Nagle & Zhou (2024)
doi:10.2139/ssrn.4693148



The UC research-to-infrastructure pipeline

Data professionals built open data infrastructure: FAIR, DMPs, curation, provenance.

FAIR4RS is the same inflection point for software — and UC has been showing the way for 50 years.

BSD Unix · Ceph · Apache Spark · Jupyter · RISC-V · Ray

The pipeline from university research to global infrastructure is not rare or accidental.

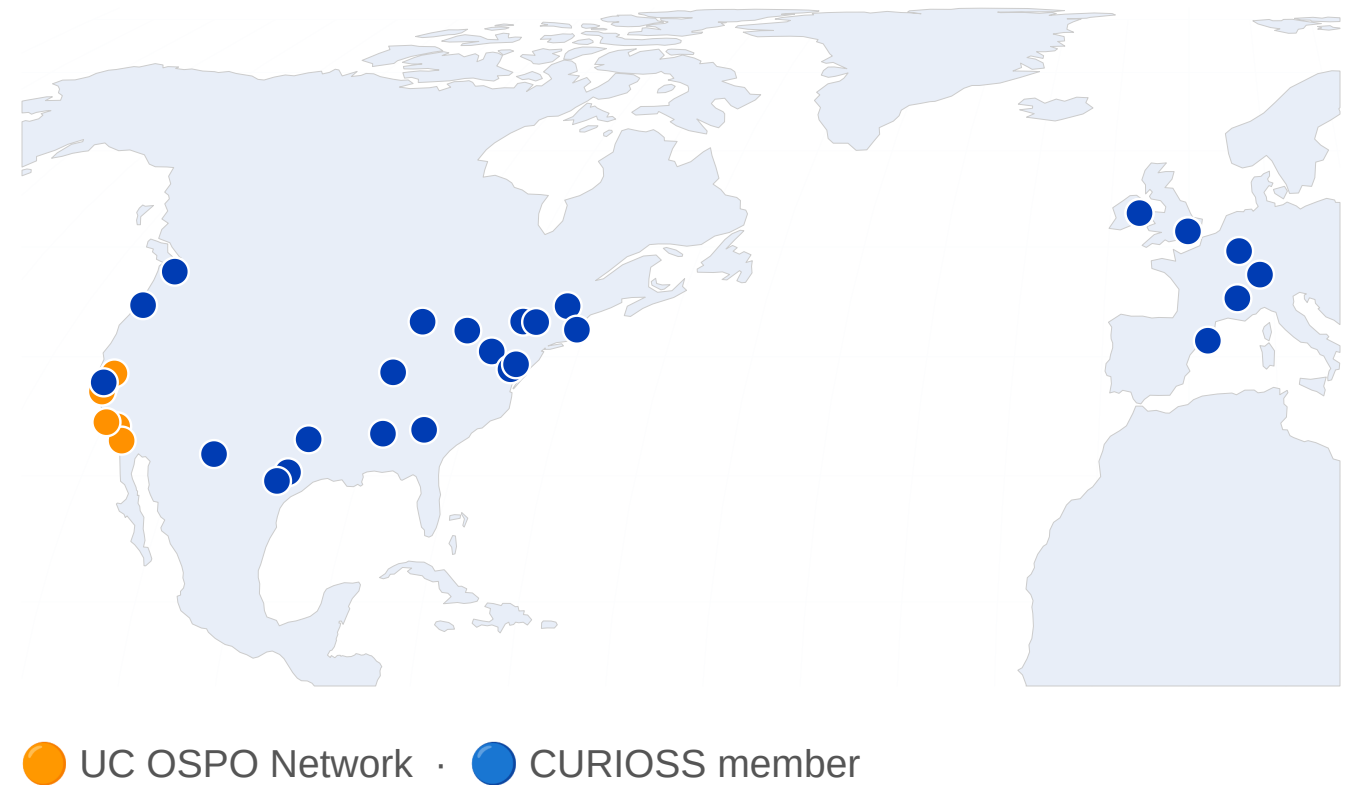
Barker et al. (2022) doi:10.1038/s41597-022-01710-x



What is an academic OSPO?

An **Open Source Program Office** supports, governs, and promotes open source within an institution.

- Originated in industry: ~77% of large tech companies have one
- First academic OSPO: Johns Hopkins, 2019
- **34 academic OSPOs** in CURIOSS worldwide and growing
- Core functions: policy, licensing guidance, best practices, education, community



The UC OSPO Network



Discovery · Sustainability · Education



Discovery

Mapping who does what across the UC system

- 200,000+ repos scanned; ~52,000 institutionally affiliated
- 294-respondent system-wide survey
- UC Open Repository Browser (UC ORB)



Sustainability

Keeping projects and communities healthy

- **58%** of UC contributors are also maintainers
- Cross-campus licensing working group (UCOP + tech transfer)
- Project health assessment (OSSPREY)



Education

Coordinated training across campuses

- Gap analysis → curriculum inventory → learning pathways
- Published at ucospo.net/education
- Coordinated with Carpentries infrastructure



Lessons from two years

What the network model enables that single campuses cannot:

1. **Shared staffing:** community manager, licensing specialist, technical roles
2. **Policy leverage:** the network has standing with UCOP; individual campuses do not
3. **Data at scale:** the GitHub and survey analyses require system-wide scope to mean anything
4. **Equity:** smaller campuses get services they could not build alone
5. **A replicable model:** \$1.85M produced something other state systems can follow

Each campus contributes unique expertise; the network routes it to where it's needed.



Where this meets your work

For data professionals, the OSPO sits at the intersection of:

- Research software engineering ↔ data management
- Licensing compliance ↔ open data policy
- Reproducibility tooling ↔ FAIR / FAIR4RS
- Training infrastructure ↔ data literacy programs
- RSE community ↔ library research support

For your institution:

- Where does open source software governance currently live?
- What would it take to treat it as infrastructure rather than a side project?



You're already doing this work



Learn more / connect

UC OSPO Network: ucospo.net

- Education site: ucospo.net/education
- Mailing list, Slack, events calendar

Global network: curiooss.org, 34 academic OSPOs worldwide and growing

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Slides: tinyurl.com/iassist2026



Handout: tinyurl.com/iassist2026-handout



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References

- Barker, M., Chue Hong, N. P., Katz, D. S., et al. (2022). Introducing the FAIR Principles for research software. *Scientific Data*, 9, 622. doi:10.1038/s41597-022-01710-x
- Gomez, J., Lovell, E., Lieggi, S., Cardenas, A. A., & Davis, J. (2025). Recipe for discovery: A pipeline for institutional open source activity. arXiv:2506.18359
- Hettrick, S. et al. (2014, updated 2022). *Research Software Survey*. Software Sustainability Institute. doi:10.5281/zenodo.14809
- Hoffmann, M., Nagle, F., & Zhou, Y. (2024). The value of open source software. Harvard Business School Working Paper 24-038. doi:10.2139/ssrn.4693148
- Ruff, N. (2026). The role of foundations in advancing open collaboration and innovation. *UC Open Summit*. youtu.be/eBriL3CDNeo
- Scarlett, V., Curty, R. G., Gomez, J., Langdon, L., Janée, G., & Budden, A. E. (2025). A system-wide snapshot: A multi-campus survey of open source contributors at the University of California. SocArXiv. doi:10.31235/osf.io/p8bx6_v1
- Tidelift. (2024). *The 2024 state of the open source maintainer report*. Tidelift.

